

P2 Neuro/MSK Practice Questions

Week 8

Questions

1. Which of the following statements is TRUE regarding whiplash associated disorders?
 - A) Most whiplash injuries eventually require surgical interventions
 - B) It is usually treated with neck immobilization with a C-collar
 - C) It is common to have spinal cord injury associated with whiplash injuries
 - D) It is a common injury during multi-vehicle collisions
2. A patient is complaining about pain in the posterior left thigh and imaging reveals a left paracentral disc herniation at L5/S1 level. Which nerve root is most likely affected?
 - A) L4
 - B) L5
 - C) S1
 - D) S2
3. Which of the following is NOT a typically radiographical finding for degenerative shoulder arthritis?
 - A) Non-uniform narrowing of joint space
 - B) Cyst
 - C) Osteophytes
 - D) Erosion
4. A patient comes into your office after falling on their elbow while playing sports, now complaining about numbness and tingling with their pinky. What spinal roots is the affected nerve originated from?
 - A) C5-T1
 - B) C5-C7
 - C) C8-T1
 - D) C7
5. Which of the following is TRUE about rotator cuff disease classification and management?
 - A) Stage I is indicated by a partial thickness tear
 - B) Stage II is indicated by a full thickness tear
 - C) Stage III can be treated with physiotherapy
 - D) Stage IV is usually treated with arthroplasty
6. Which of the following physical exam findings is most consistent with bicipital tendonopathy?
 - A) Postive Speed's test
 - B) Postive Hawkin's Test
 - C) Positive Neer's test
 - D) Negative Yergasen's test
7. A patient has recently suffered from a mid-shaft humorous fracture. Which of the following complications is the LEAST concerned of?
 - A) Malunion of humerus
 - B) Nonunion of humerus

- C) Radial nerve injury
 - D) Median nerve injury
8. Which of the following statements is FALSE regarding elbow dislocation?
- A) Anterior dislocation is more common than posterior dislocation
 - B) The most common mechanism of the injury is falling onto outstretched hand
 - C) Ulnar nerve is most of concern during a typical elbow dislocation
 - D) One of the complications to watch for is the compartment syndrome of the forearm
9. Which of the following statements is TRUE regarding medial epicondylitis and lateral epicondylitis?
- A) Medial epicondylitis is the tendinopathy at the common extensor origin
 - B) Lateral epicondylitis can produce pain with passive stretch of wrist extension
 - C) Medial epicondylitis can produce pain with resisted wrist extension
 - D) Lateral epicondylitis is more common with the age group of 30's-40's

10) A 35-year old patient presents to your office with right-sided back pain mainly. The pain radiates to the posterior thigh. It gets worse with extension and is relieved with flexion. The patient has no sensory symptoms or other red flags. What is the most likely diagnosis?

- A) L5 radiculopathy
- B) Facteogenic pain in the lumbar spine
- C) Facetogenic pain the sacral spine
- D) Cauda equina syndrome
- E) None of the above

11) During your ER rotation, a patient is brought to the ER complaining of a sudden loss of sensation in the buttocks area. They've been experiencing bilateral leg and back pain for the past 24 hours. What is the best next step?

- A) Reassure that the patient that this is mechanical back pain and prescribe NSAIDs.
- B) Refer to physiotherapy
- C) Order an emergency MRI
- D) None of the above

12) A 45-year old patient presents to your clinic with left shoulder stiffness that started suddenly 2 weeks ago. After doing a thorough physical exam, you observe that the patient has a globally restricted range of motion. The patient has T2DM but is otherwise healthy with no previous medical history. What is the most likely diagnosis?

- A) Adhesive capsulitis
- B) Osteoarthritis
- C) Shoulder dislocation
- D) None of the above

13) You perform a physical exam on a patient has been complaining of progressive shoulder pain and they report a positive painful arc test. Which muscle is most likely implicated?

- A) Infraspinatus
- B) Teres minor
- C) Both A and B
- D) Supraspinatus
- E) Both A and D

14) A 35-year-old professional athlete presents to your office with anterior shoulder pain that gets worse with shoulder flexion and elbow flexion. The patient has no sensory symptoms. Which structure is most likely impinged?

- A) Biceps brachii (short head tendon)
- B) Biceps brachii (long head tendon)
- C) Musculocutaneous nerve
- D) Both A and B
- E) None of the above

15) Following a motor vehicle accident, a patient is unable to extend their arm or forearm. The patient is also unable to fully abduct their shoulder. Which of the following structures has most likely been damaged?

- A) Lateral cord
- B) Posterior cord
- C) Medial cord
- D) Ansa cervicalis

16) A 25-year-old got injured during a soccer game. The patient experiences significant pain and is unable to fully abduct their shoulder. X-ray of the shoulder joint shows damage to the proximal humerus. Which of the following structures has most likely been damaged?

- A) Axillary nerve
- B) Anterior circumflex humeral artery
- C) Posterior circumflex humeral artery
- D) A and B
- E) A and C

17) A few weeks after falling on their elbow, your patient presents with a hot, swollen and tender elbow. What is the most appropriate next step?

- A) Aspirate the joint to rule out Olecranon bursitis
- B) Reassure the patient and prescribe NSAIDs for pain
- C) Order an MRI of the joint
- D) None of the above

18) A patient presents to your office with pain in their elbow. On physical exam, the pain gets worse with resisted elbow extension and resisted middle finger extension. What is the most likely diagnosis?

- A) Lateral epicondylitis
- B) Triceps tendinopathy
- C) Medial epicondylitis
- D) None of the above

Answers

1. D - Most whiplash injuries have muscles involved rather than the spinal cord and can be managed with conservative management with simple neck mobilization.
2. C.
3. D - Erosions are typically found in inflammatory arthritis. Asymmetric joint space narrowing, osteophytes, cysts and subchondral sclerosis are the typical radiological findings for degenerative (non-inflammatory) arthritis.
4. C.
5. D - Stage I: no structural defects; II: partial tear; III: full tear.
6. A - Positive Speed's and Yergason's tests indicate bicipital tenosynovitis. Positive Neer's and Hawkins's tests indicate rotator cuff impingement.
7. D - Radial nerve is more prone to injury during humerus fracture due to its close proximity to the bone. Malunion and nonunion are the common complications that should be watched for with all fractures.
8. A - Posterior dislocations are the most common.
9. D - Medial epicondylitis is at the common flexor origin and can produce pain with passive stretch of wrist extension, and resisted wrist flexion; lateral epicondylitis is at the common extensor origin and can produce pain with passive stretch of wrist flexion, and resisted wrist extension.
10. B. This presentation (pain that is aggravated by extension and relieved with flexion) is consistent with facetogenic pain of the lumbar spine
11. C. An MRI is needed to rule out cauda equina syndrome
12. A. Given the globally restricted range of motion and the patient's history of T2DM, adhesive capsulitis (frozen shoulder) is the most likely diagnosis here
13. D. A positive painful arc test is suggestive of supraspinatus impingement
14. B. This presentation is most consistent with proximal biceps tendinopathy in which the tendon of the long head of the biceps becomes impinged.
15. B. The symptoms presented here suggest injury to the posterior cord from which both the axillary and radial nerves originate.
16. E. An injury to the proximal humerus can cause damage to the surgical neck. The axillary nerve and the posterior circumflex humeral artery both course through here.
17. A. Given the inflammatory symptoms, aspirating the joint would be the most appropriate next step here.
18. A. The scenario presented here is suggestive of a positive test for lateral epicondylitis

Week 9

Questions

1. A patient presents to clinic with loss of sensation over palmar aspect of digit 2 and 3 of right hand. The patient has a history of repetitive strain work as a factory worker and has a positive tunnel sign upon percussion of carpal tunnel. This leads you to suspect median nerve neuropathy. However, you are a smart medical student and know that a C6 radiculopathy is also on the differential for this patient. Which of the following clinical findings, if present, would be more consistent with a C6 radiculopathy and causes you to question your initial diagnosis of median nerve neuropathy.

- A. The patient reports that the area of sensory loss over the palmar aspect of digit 2 and 3 is well defined
- B. Weakness of biceps flexion and wrist extension
- C. Wasting of thenar muscles and inability to oppose thumb
- D. Loss of dexterity of 2nd and 3rd digit

2. Which of the following statements about the ulnar nerve is false

- A. The most common site on injury for the ulnar nerve is at the medial epicondyle of the elbow
- B. Deep branch of the ulnar nerve innervates the hypothenar muscles, dorsal and palmar interossei, and deep head flexor pollicis brevis
- C. Damage to the ulnar nerve may result in loss of sensation on the palmar and dorsal aspect of the lateral 1.5 digits
- D. Ulnar nerve also runs with ulnar artery in Guyon's canal

3. Which of the following statements with regards to neurons and nerve injuries is true?

- A. Endoneurium is the innermost layer of connective tissue within a nerve and it directly surrounds each axon
- B. Peripheral nerves consist of a single axon wrapped in three layers of connective tissue, which are (from outer to inner most), epineurium, perineurium, and endoneurium.
- C. Neurotmesis nerve injuries have the ability for spontaneous recovery because the epineurium is still intact, which serves as a path for regenerating fibers to grow along
- D. Myelin injuries take a relatively longer time to recover than axonal injuries, possessing a recovery rate of 1 inch per month (1mm/day)

4. Which of the following pathologies is it NOT possible for electrodiagnostic studies (EDX) (included EMG and NCS studies) to detect?

- A. Axonal injuries
- B. Myelin injuries
- C. Myopathies
- D. Central Nervous System Lesions

5. A patient is to undergo coronary bypass surgery and needs a graft from another blood vessel from another site in the body to complete the procedure. You select the Great Saphenous Vein to be the graft. What nerve are you likely to damage when you go to harvest this graft and what would the sensory neurological deficits be?

- A. Saphenous nerve: loss of sensation along the medial thigh
- B. Saphenous nerve: loss of sensation over medial knee, leg, and foot
- C. Obturator nerve: loss of sensation over medial thigh
- D. Obturator nerve: loss of sensation over medial knee, leg, and foot

6. A 76 year old female patient with osteoporosis presents to the ER in immense pain over her right hip around the region of the greater trochanter for the past 6hrs. She reports rolling off her couch when she was napping and landing on her hip. You order an xray and confirm an intracapsular femoral neck fracture. What is the next step in the management of this patient?

- A. Recommend conservative management for this patient – non-weight bearing for 3-4 weeks
- B. Surgical arthroplasty within 2-4 weeks
- C. Urgent open reduction and internal fixation (ORIF) within 48hrs
- D. Urgent hemiarthroplasty or total hip arthroplasty within 48hrs

7. Which of the following is NOT a feature of hip osteoarthritis?

- A. Presence of osteophytes and narrowing of joint space on xray is required for diagnosis
- B. Pain is worse with prolonged activity
- C. When asked to identify the pain, patient may grip the affected hip just above the greater trochanter between the thumb and index finger, in what is known as a positive “C” sign
- D. Most commonly presents in adults over 50 and classically affects more females than males

8. Jon Hamilton, a 19-year-old man comes to the ER because of right wrist pain and swelling 2 hours after falling on an outstretched hand while skateboarding trying to impress his Ram Ranch™

roommates Aaron and Devin. His right wrist is swollen and tender; range of motion is limited by pain. There is tenderness to palpation in the area between the tendons of the abductor pollicis longus, extensor pollicis brevis, and extensor pollicis longus muscle. His thumb can be opposed actively toward the other fingers. Muscle strength of the right hand is decreased.

Which of the following is the most likely diagnosis?

- A) Colles fracture
- B) Transscaphoid perilunate dislocation
- C) Trapezium fracture
- D) Scaphoid fracture

9. Liam LeClair, a 24-year-old man, comes to the walk-in clinic with Right shoulder pain after an innertube waterpolo game. He states he heard a “pop” and then immediately felt 8/10 pain when an absolute bomb was blocked in his hand by Bonnie while he was throwing at team Lee’s goalie in the last 15 seconds of the game. He is stable, and sitting stoically in the exam room. Upon inspection his right shoulder appears slightly abducted and externally rotated. The acromion appears prominent, and there is loss of the normal rounded appearance of the shoulder (see image on R).



He resists all movement due to pain. You perform an assessment of his right arm’s neurovascular status, which nerve are you most worried about ?

- A) Radial nerve
- B) Musculocutaneous nerve
- C) Axillary nerve
- D) Ulnar nerve

10. You ordered bloodwork for a patient, the phlebotomist is preparing to draw blood from their arm. Which vein is the most likely spot for venous access?

- a. Cephalic vein
- b. Median cubital vein
- c. Greater saphenous vein
- d. Dorsal venous network

11. A 28-year-old woman, gravida 2, para 1, at 24 weeks' gestation comes to the physician for a prenatal visit. She reports dull aching pain and paresthesia over her left hand for the last few weeks. The pain radiates to her shoulder and is worse at night. Her hand feels numb upon waking up in the morning. She has a sister who has multiple sclerosis. Her current medications include iron supplements and a multivitamin. Vital signs are within normal limits. When the wrist is passively held in full flexion, aggravation of paresthesia is perceived immediately. Which of the following is the most likely explanation for this patient's symptoms?

- a. Demyelination of her peripheral nerves

- b. Cervical radiculopathy
- c. Median nerve compression
- d. Lower trunk palsy of the brachial plexus

12. One day while preparing dinner, Mohamed Aly a second year medical student suffers a cut to the thenar aspect of his left hand while cutting vegetables. After the initial bleeding is controlled he notices that sensation over the thenar aspect and central palm is decreased. Which nerve did he injure?

- a. Palmar cutaneous branch of the median nerve
- b. Digital branch of the median nerve
- c. Radial cutaneous nerve
- d. Pollicis cutaneous nerve

13. One day while bull riding at the Ram Ranch™ cowboy and second year med student, Adam Levschuk, suffered a horn injury to his femoral triangle from the medial side. In what order would the structures be in danger from the bull's horn?

- a. Lymphatics, Nerve, Artery, Vein
- b. Lymphatics, Vein, Artery, Nerve
- c. Vein, Artery, Nerve, Lymphatics
- d. Nerve, Artery, Vein, Lymphatics

14. While being arrested by London's finest for public indecency after a night at Jack's. James Bunker's arms were handcuffed behind his back. What actions did his upper limb have to perform to get to this position? Write your answer here:

Answers

1. B. Biceps and wrist flexion are part of the C6 myotome which would be impaired if there was injury at the C6 nerve root as would be caused by a radiculopathy. All the rest are findings that would be consistent with a median nerve neuropathy at the wrist (more suggestive here due to positive tinell sign at the carpal tunnel area).
2. C. The ulnar digits are the MEDIAL 1.5 digits, not lateral
3. A. B is false, there are many axons contained in a single nerve. C is false, neurotmesis injuries do not have the ability to spontaneously recover because the epineurium is not intact. D is false, axonal injuries are slower with a rate of 1inch/month.
4. D. It is not possible for EDX studies to detect CNS pathology like stroke or multiple sclerosis. All the others are possible. NCS can be used to assess health of myelin sheath and axon while EMG can be used to measure muscle functions in isolation.
5. B. Saphenous nerve runs in close proximity to great saphenous vein and supplies the anteromedial aspect of the knee, leg, and foot. Obturator nerve runs deep in the medial

compartment of the thigh and is unlikely to be damaged during this procedure, as the great saphenous vein is a superficial vein.

6. D. With hip fractures, surgery outcomes are best when surgery is performed within 24-48hrs of the fracture. For older patients (60+), the surgical method of choice is a hemiarthroplasty or total hip arthroplasty over ORIF.
7. A. These changes may be present on Xray and may be used to support the diagnosis of hip OA, however, it is possible to make this diagnosis on the basis of clinical presentation alone. Thus they are not required for diagnosis. All the other statements are true.
8. **D - Scaphoid fractures** typically occur following a fall on an outstretched hand. If pain occurs in the anatomical snuffbox (area between the tendons of the abductor pollicis longus, extensor pollicis brevis, and extensor pollicis longus muscle) after trauma, the injury should be treated as a scaphoid fracture until proven otherwise to prevent possible complications such as avascular necrosis and bone nonunion.
9. **C - The Axillary nerve** is most commonly damaged nerve in an anterior shoulder dislocation.
10. B. Median cubital vein is the most common venous access spot for phlebotomy
11. C. Median nerve compression would be the cause of her symptoms given the clinical picture of carpal tunnel syndrome.
12. A. The palmar branch of the median nerve can be commonly injured with cuts to the thenar aspect of the palm.
13. B. NAVEL is the mnemonic from Lateral to medial. So from MEDIAL to Lateral it would be LEVAN
14. Shoulder **aB**duction, extension, Internal rotation, and Elbow flexion.

Week 10

Questions

1. 5 year old girl fell on the playground. Xray showed a fracture in the proximal humerus, 100% displacement in the plane of the joint. What would be the preferred treatment?
 - E) Put on a sling
 - F) Physiotherapy
 - G) Closed reduction and casting
 - H) Emergency surgery
2. Erin is a 8 year old girl presenting with thoracic scoliosis angled at 23°. What are the possible treatment strategies for her? (select all that apply)
 - A) Education and monitoring
 - B) Spine orthosis
 - C) NSAIDS for pain
 - D) Surgical intervention
3. Patient with a family history of neurofibromatosis presents with a mass that is less than 5cm, tethered, and deep on the hip. What is the best sampling method for this lesion?
 - A) FNAB
 - B) Core biopsy
 - C) Incisional biopsy
 - D) Excisional biopsy
4. The biopsy sample confirms sarcoma. What are the next management steps?
 - A) Pre op chemo
 - B) Pre op radiation
 - C) Surgery
 - D) Adjuvant chemo
 - E) Post op chest CT
5. Why is it important to have multiple Xray views when assessing a fracture? (select all that apply)
 - A) To pick up microfractures that may only be seen in one view
 - B) To differentiate between a transverse and linear fracture
 - C) To assess whether an oblique fracture is actually a spiral fracture
 - D) To check whether tendons are involved in an avulsion fracture
 - E) To confirm a rotational fracture
6. Which of the following is true for fracture healing?
 - A) Primary healing is healing that comes spontaneously and seen more commonly
 - B) Bony callus forms at fracture site, then revascularization occurs
 - C) Distal humerus takes longer to heal than proximal humerus
 - D) Malunion refers to no healing in 6-9 months
 - E) Healing takes longer if the fracture is extraarticular since there is no synovial fluid involved

7. 50 year old patient presents to the emergency department following a motor vehicle accident. Xrays show that there is a talar neck fracture. The patient undergoes ORIF. What complication will the patient most likely experience following this injury?
- A) Posttraumatic arthritis
 - B) Osteonecrosis
 - C) Osteoarthritis
 - D) Heterotopic ossification
8. Patient presents with an open tibial fracture with a skin defect measuring 5cm. Which of the following are true? (select all that apply)
- a. There is an increased chance of nonunion and infection
 - b. Obtain tetanus status and administer tetanus prophylaxis
 - c. Take a swab and wait for culture results before administering antibiotics
 - d. Contact plastic surgery to perform flap surgery
9. You are in the emergency room assessing Rachel Green, a 31 year old female. Rachel was driving her friend's Porsche at high speed on the 401 when the car in front of her braked suddenly. Rachel collided into the car in front of her. On your exam, you notice Rachel has some mild knee pain, minimal swelling of her right knee and decreased range of motion. Given the mechanism of injury, what knee ligament did Rachel most likely injure?
- a. Medial collateral ligament
 - b. Lateral collateral ligament
 - c. Anterior cruciate ligament
 - d. Posterior cruciate ligament
10. Tobiloba is a 19 year old woman coming to see you in the family medicine clinic with gradually worsening bilateral anterior knee pain. She describes the pain as deep and aching. She first noticed the pain when she increased her hurdles training volume, and initially the pain was brought on by exercise. However, the pain is now brought on when she climbs stairs and stands up after prolonged periods of sitting. Based on the most likely diagnosis, what would you offer Tobiloba? Choose all that are correct.
- a. Physiotherapy
 - b. Knee arthroscopy
 - c. Activity modifications
 - d. Bracing
 - e. Taping
11. You are a family medicine resident seeing Kavi, 58 years old, who complains to you about pain in their legs. They have aching pain in both legs that sometimes feels itchy. Kavi works in customer service and finds the pain to be worse after a long shift standing. On physical exam, you see dilated, tortuous veins on both legs. Which of the following is not recommended for Kavi?

- a. Daily exercise
 - b. Compression stockings
 - c. Daily hot water soak
 - d. Laser therapy
 - e. Daily elevation of the legs
12. Milica is a 30 year old woman in your family medicine clinic seeking help for pain on the dorsum of her right foot which radiates to the space between her first and second toes. She recently took on a job at a ski resort where she teaches ski lessons for six hours a day. Which nerve has Milica most likely compressed?
- a. Tibial nerve
 - b. Common peroneal nerve
 - c. Superficial peroneal nerve
 - d. Deep peroneal nerve
13. How does physical activity affect the cardiovascular system? Check all that are correct.
- a. Increases oxygen uptake
 - b. Decreases blood pressure during exercise
 - c. Increases blood viscosity
 - d. Improves endothelial function
14. Akbar is a 56 year old patient interested in adding physical activity to his lifestyle. He comes to you for advice because he is fairly sedentary and has a history of hypertension, with average readings of 185/100. Which of the following would you NOT recommend?
- a. Isometric exercises
 - b. An ACE inhibitor
 - c. Prolonged warmup and cool-down
 - d. At least one rest day per week
 - e. Calcium channel blockers
15. Blanche, a 55 year old woman, comes to your walk-in clinic seeking help for pain on the bottom of her foot. The pain localizes to her heel and she tells you the pain is the worst when she gets out of bed in the morning and when she goes for a run. Based on your suspected diagnosis, which of the following would you not recommend for Blanche?
- a. Night splints
 - b. Shoes with arch support
 - c. Physiotherapy
 - d. Corticosteroid injections
 - e. Cushioned heel pad
16. Pes planus commonly occurs after an injury to which of the following muscles?
- a. Tibialis posterior
 - b. Quadratus plantae

- c. Peroneus longus
- d. Tibialis anterior

Answers

1. A - young kid, so lots of ability to remodel naturally, only needs slight traction using a sling. There are positive factors for bone remodeling (more than 2 years of growth remaining, defect in plane of joint)
2. A, B, D - consider surgery since she has multiple risk factors for progression (girl, premenarchal, thoracic spine). Pain is rarely a presenting feature for scoliosis.
3. B - suspect sarcoma, so best sampling method is core biopsy. FNAB cannot provide cell organization and structure. Incisional is more invasive than core biopsy. Excisional biopsy is not ideal for sarcoma.
4. B, C, E - treatment is pre op radiation, then surgery. These tumors are resistant to chemotherapy, so no chemo done. Post op chest CT important to monitor recovery.
5. A, C, E - some microfractures may only be seen in one view. A transverse and linear fracture are very different, and can be differentiated from a single view. A spiral fracture may look like an oblique fracture in one view, so it is important to obtain another view to differentiate the two. Tendons cannot be seen well on Xrays. Rotational fractures usually need two views to look at joints proximally and distally to interpret the degree and type of rotation.
6. C - A is wrong because this describes secondary healing, primary healing is the contact healing after closed reduction or ORIF. B is wrong because revascularization occurs first, then formation of bony callus occurs after. C is correct. D describes nonunion. E is wrong because intraarticular fractures take a longer time to heal.
7. A - posttraumatic arthritis is the most likely complication following ankle fracture. Osteonecrosis is possible, but not as likely as posttraumatic arthritis. Osteoarthritis and heterotopic ossification are not likely in the ankle.
8. A, B - there is an increased chance of complications (infection, nonunion, time to union) for open fractures. It is important to get tetanus status and prophylaxis for all open fractures. Do not wait for culture to administer antibiotics. Since this is a grade 2 open fracture, there is no need for flap surgery.
9. D. This is a classic dashboard injury causing the tibia to move posteriorly relative to the femur when the knee is flexed. Patients typically present with poorly defined pain and minimal swelling.
10. A, C, D, E. Given the presentation and the demographics of this case, Tobiloba most likely has patellofemoral pain syndrome (PFPS). Because PFPS is related to increased Q angle, physiotherapy may focus on quadriceps strengthening (especially vastus medialis) to prevent lateral translation of the patella during knee flexion. Knee surgery is not indicated. Activity modifications are indicated, and bracing and taping may also help prevent lateral patellar translation.

11. ABDE. Kavi has varicose veins. Exercise, compression stockings, and daily elevation of the legs can help return the blood towards the heart. Laser therapy damages the varicose veins, causing formation of scar tissue and eventual disappearance of the vein. Daily hot water soaks are NOT recommended as they can further vasodilation and venous stasis.
12. D. Tight boots such as ski boots can compress the deep fibular nerve where it passes under the extensor retinaculum.
13. A & D
14. A. ACE inhibitor or calcium channel blocker will help control Akbar's hypertension to make exercise safe. Including a warmup of 15minutes and a cool-down of 10minutes is an important consideration for patients with hypertension. One rest day per week will help Akbar avoid injury. Isometric exercises, overhead exercises and supine exercises are not recommended for patients with hypertension.
15. Blanche most likely has plantar fasciitis. While A, B, C, and E are important non-pharmacological interventions, corticosteroid injections do not play a role.
16. A. While peroneus longus and tibialis anterior also play a role in supporting the medial longitudinal arch, the main contributor is tibialis posterior, and thus injury to tibialis posterior is a common cause of pes planus.

Week 11

Questions

1. Match the rheumatologic condition (a,b,c,d,e,f) to its *best* corresponding investigation (i, ii, iii, iv, v, vi). Bolded are some unique investigations for help in answering this question!
 - a. Systemic lupus erythematosus
 - b. Inflammatory myositis
 - c. Sjogren's
 - d. Scleroderma
 - e. ANCA vasculitis
 - f. Enthesitis-related arthritis
 - I. CBC, ANA, ESR, RF, **anti-Ro** (aka SSA), **anti-La** (aka SSB)
 - II. ANA nucleolar pattern, **Scl70** (topoisomerase 1)
 - III. **CK**, ESR, ANA
 - IV. ANA, then **anti-DNA** (if ANA is positive), anti-Smith
 - V. **HLA B27**, ESR, CRP, CBC
 - VI. **ANCA**, PR3, MPO
2. In which of the following cases is ordering an ANA indicated? Choose 1
 - a. A family doctor is unsure if a patient has an autoimmune disorder and the patient had a negative ANA result last year (no new symptoms).
 - b. A rheumatologist wants to follow a SLE patient's progress/response to treatment.
 - c. A patient presents to the ER with delirium, acute pericarditis, fever, alopecia, and proteinuria.
 - d. A patient is presenting with swollen and achy knees that are stiff (<30 min every morning), prone to night pain, and have a reduced ROM.
3. A 63 y.o Finnish patient presents to the ER with sudden painless vision loss, an awful headache, pain with chewing, and pain with laying his head on their pillow at night. He also mentions pain in his shoulders and hips and classic B symptoms (fever, weight loss, night sweats)! What should the next steps be? Choose 3
 - a. Give high-dose IV methylprednisolone
 - b. Give medium dose prednisone.
 - c. Give low-dose prednisone (5 mg)
 - d. Consult oncology urgently due to risk of intracranial malignancy
 - e. Obtain ESR and CRP
 - f. Consult ophthalmology for a temporal artery biopsy
4. Which of the following is false?
 - a. Prior to diagnosing primary vasculitis, serum protein electrophoresis, urine toxicology screens, sputum cultures, Hep B+C serology, and HIV testing must be taken.

- b. Common signs of primary vasculitis includes ecchymosis, intracranial hemorrhages, cranial nerve neuropathies, and concurrent infections.
- c. Giant cell arteritis cannot affect people who are 49 years old or younger.
- d. A patient with ANCA vasculitis can still have a negative ANCA.
- e. Patients with primary vasculitis need to have the following investigations to check for organ involvement: ALT, albumin, Cr, urinalysis, chest x-ray.
- f. Vasculitis involves blood vessel thickening from inflammation and a resulting turbulent blood flow.

5. Match the complication (a,b,c,d,e) to the condition (i, ii, iii, iv). Warning: there may be repeats (e.g. one condition has 1+ complications)

- a. Aspirations, arrhythmia, adenocarcinomas
- b. MALT lymphoma
- c. Hypertension, atherosclerosis, diabetes and strokes
- d. Interstitial lung disease
- e. Renal crisis

- I. SLE
- II. Scleroderma
- III. inflammatory myositis
- IV. Sjogren's

6. Based on the 15% rule, which comorbidities will affect only 15% of scleroderma patients. Choose 3:

- a. Raynaud's phenomenon
- b. GERD
- c. Dysphagia
- d. Sjogren's
- e. Current digital ulcers
- f. Pulmonary arterial hypertension

7. Match the common presentation (a,b,c,d) to the pediatric condition (i, ii, iii, iv).

- a. A child (between 3-10 y.o) who was sick with a URTI about a week ago now has sudden unilateral hip pain and their hip is held in slight FABER position
- b. An adolescent athlete (8-14 y.o female, 10-15 y.o male) has a painful bump over their tibial tuberosity and xray shows a teardrop sign.
- c. An obese adolescent (~ 12 y.o female, 14 y.o male) has gradual left hip pain that refers to the knee, antalgic gait, and can't weight bear
- d. A child (between 4-8 y.o) who has a low body weight and protein S/C deficiency has gradual vague hip pain and stiffness and limb-length discrepancy

- I. Transient synovitis
- II. Leg calve perthes
- III. Osgood-Schlatter

IV. Slipped capital femoral epiphysis

8. Which of the following is incorrect about pediatric conditions related to limp?
 - a. The main treatment for avascular necrosis of the proximal femoral head (epiphysis) is close observation (along with activity restriction and NSAIDs/Tylenol) because it will resolve on its own.
 - b. Slipped capital femoral epiphysis involves the femur shifting posteriorly and inferiorly.
 - c. Transient synovitis requires non-urgent xray and hip ultrasound for diagnosis.
 - d. Males are more affected than females for leg calve perthes, slipped capital femoral epiphysis AND transient synovitis.
9. Over the phone, your friend Marnie tells you that she is sad because her manicurist made mean comments about her nails. Some insults include: "Your nails have pits and ridges, yellow dots and dark streaks, and are basically lifting right off. Your fingers also look like sausages." Your friend is also stressed because she thinks she has relentless dandruff, pinkeye, and arthritis in her knees. You suspect your friend actually has a type of spondyloarthritis. Which is it?
 - a. Reactive arthritis
 - b. Enteropathic arthritis
 - c. Ankylosing spondylitis
 - d. Psoriatic arthritis
 - e. Sacroiliitis
10. Beside giving emotional support, you want to help Marnie in some way. You recommend all of the following csDMARDs (conventional synthetic disease-modifying antirheumatic drugs) but tell her to avoid which one?
 - a. Methotrexate
 - b. Sulfasalazine
 - c. Hydroxychloroquine
 - d. Leflunomide
11. You are meeting with your clerk friend Osler. For the past year, he has had unrelenting headaches, forgets things easily, and he never feels refreshed no matter how much he sleeps. You joke that he might be overworked and just needs a break, but he says that he also has severe chest pain, jaw pain, back pain, bilateral shoulder pain, leg pain, abdomen pain, AND hip pain. None of his doctors could explain his symptoms. You suspect he has a disorder relating to his descending (inhibitory) and ascending (facilitatory) pain pathways resulting in decreased serotonin and increased substance P and glutamate. This time you recommend all of the following treatments except which one?
 - a. Yoga and mindfulness techniques
 - b. Pregabalin
 - c. Amitriptyline
 - d. Prednisone
12. Which of the following cases likely has polymyalgia rheumatica?
 - a. A 64 y.o patient who has a history of giant cell arteritis presents with new weight loss and also pain in shoulder and hip.

- b. A 43 y.o male presenting with pain and stiffness in the shoulder and hip girdles with profound morning stiffness and fatigue.
 - c. A 56 y.o febrile female presenting with pain in just her left shoulder, which is also warm and swollen.
 - d. A 73 y.o patient with long-standing pain in hip (which is not stiff), worse with activity. Her shoulder muscles has weakness but no pain. They sadly have not responded to prednisone that you previously prescribed.
13. Which of the following are not characteristics of cellulitis?
- a. Infection on palmar surface of the hand
 - b. Superficial skin infection
 - c. Treatments generally require rest, elevation and antibiotics
 - d. Margins are irregular
14. What is the best investigation method when you suspect osteomyelitis in a patient?
- a. Bloodwork - Cr
 - b. Bloodwork - WBC
 - c. CT scan
 - d. MRI
15. Your 45 year old male patient comes in for regular blood work. Results show that he has elevated uric acid. What are your next steps? Choose the best answer.
- a. Start allopurinol
 - b. Advise him to start eating more seafood
 - c. Reassurance
 - d. Aspirate his knee joint
16. Which of the following - if found in your patient highly suspicious of pseudogout - does not necessarily support the diagnosis of CPPD? (Choose multiple answers)
- a. Chondrocalcinosis (deposition of calcium pyrophosphate) on imaging
 - b. Osteoarthritis
 - c. High uric acid level
 - d. Dehydration
 - e. Attack lasting 7-10 days
17. Which of the following are not appropriate treatment plan for CPPD?
- a. Colchicine
 - b. Intra-articular corticosteroid injection
 - c. COXIB
 - d. Prednisone (oral)
18. True or False: X-ray is the best way to diagnose chronic osteoarthritis

- a. True
- b. False

19. Your 65 year old patient has early stage Osteoarthritis. She strongly asks you if she could get a joint replacement surgery. What would be your response?

- a. Sure! Let's go ahead.
- b. No - we should perform knee arthroscopy.
- c. No - treatment for early stage OA is monitoring & reassurance with absolutely no intervention
- d. No - joint replacement surgeries are advised on end-stage OA patients who have tried every other non-invasive methods and still have failed to improve. We can instead try other methods like Tai-Chi and physiotherapy first.

20. Which of the following should we try to avoid for medical treatment of early-middle stage OA?

- a. NSAIDS - topical or oral
- b. Prednisone
- c. Intra-articular corticosteroid injection
- d. Duloxetine
- e. Tramadol

21. Which of the following affects Sacroiliac joints?

- a. Connective Tissue Disease associated arthritis
- b. Rheumatoid arthritis
- c. Seronegative arthritis
- d. Osteoarthritis

22. Which of the following favour small joints in the hand and feet, and spares distal phalangeal joints?

- a. Osteoarthritis
- b. Seronegative arthritis
- c. Infectious arthritis
- d. Rheumatoid arthritis

23. Enthesitis related arthritis is... (choose all that are correct)

- a. More common in girls
- b. Biphasic onset - at age 2-5 & at age 10-14
- c. 30% go onto develop ankylosing spondylitis
- d. HLA B27 associated (50%)
- e. Can result in sacroiliitis, dactylitis, enthesopathy and peripheral involvement

24. What type of JIA is not commonly associated with asymptomatic uveitis?

- a. Oligoarticular JIA
- b. Polyarticular JIA
- c. Enthesitis related arthritis

- d. Systemic onset JIA

Answers

1. A- iv; B - iii (anti-Jo1 and anti-PM-Scl can also be ordered to check for interstitial lung disease and scleroderma polymyositis respectively, which are disease correlations), C - i; D -ii; E - vi (PR3 stands for anti-proteinase 3 aka c-ANCA; MPO stands for anti-myeloperoxidase aka p-ANCA); F- v (30-50% of enthesitis-related arthritis have a positive HLA B27; CBC might show anemia and leukocytosis & thrombocytosis, ESR may be normal or elevated)
2. C. In answer C, the patient actually fulfills multiple clinical domains for SLE. Since an autoimmune CTD is suspected, ANA can be ordered. ANA is not ordered if there was a negative ANA in the past. It is also not ordered to follow a patient. Case D describes a patient with osteoarthritis (not inflammatory = no ANA needed).
3. A, E, F. This patient has giant cell arteritis. Low-dose prednisone is only reserved for cases where other options don't work or there are difficulty discontinuing steroids. Medium-dose prednisone is for RA flares, polymyalgia rheumatica and acute gout flare. Remember to not wait for ESR/CRP results and to start high dose corticosteroids immediately! Patients who are of Northern European descent (e.g. Finnish in this case) are at greater risk of giant cell arteritis. Intracranial malignancies need to be ruled out but doesn't require oncology consult.
4. B. Common signs of vasculitis are actually petechial/purpurial rashes and B-symptoms (infections must actually be ruled out to make a diagnosis of primary vasculitis). Clots, occlusions/ischemia and aneurysms are more of a concern than bleeding manifestations in vasculitis. A is true because infection and underlying malignancy/drugs must be ruled out.
5. A - iii, B - iv, C - i, D - i & ii, E - ii
6. DEF. Nearly all have Raynaud's and GERD and many have dysphagia.
7. A - i. B - iii, C - iv. D - ii.
8. C - transient synovitis is a clinical diagnosis based on history.
9. D - the mean manicurist's comments are the nail/finger changes found in psoriatic arthritis. The friend's dandruff may be scalp psoriatic plaques and her pink eye is conjunctivitis (found in 20% of psoriatic arthritis cases).
10. C - hydroxychloroquine is often avoided due to case reports of severe psoriasis flares.
11. D - he likely has fibromyalgia. Prednisone shouldn't be used for fibromyalgia.
12. A - giant cell arteritis often occurs with polymyalgia rheumatica. B is not correct because polymyalgia rheumatica only affects people older than 50. C is likely septic arthritis (bacterial monoarthritis). D is likely hip osteoarthritis and shoulder myositis. Polymyalgia rheumatica won't have muscle weakness and should respond dramatically to prednisone.
13. A. Cellulitis is typically present on dorsal surface of the hand.
14. D. When investigating Osteomyelitis, X-ray nor CT are sensitive enough. MRI is the most sensitive way - especially for vertebral OM. However, it's important to not routinely repeat MRI. Otherwise, NM Tagged WBC scan is also a good way to investigate OM.

15. C. Hyperuricemia does not necessarily mean gout. In this case, this patient does not have any symptoms of gout (i.e., swollen joint, red, painful). Therefore, doing further investigations and/or treatment is unnecessary. Furthermore, if you patient has gout, you should advise them to decrease the amount of seafood in their diet; and we typically do not prescribe allopurinol right away (unless it is debilitating to their life), and especially not during the gout attack. Lastly, gout is more commonly found on 1st MTP joint (big toe), not on the knee (CPPD more commonly found in knee).
16. C & E. these are characteristics that support Gout. Pseudogout typically lasts for a few days~ few weeks (takes longer to reach a peak compared to gout).
17. A. Colchicine is used to treat gout.
18. B. False - osteoarthritis is a clinical diagnosis (symptoms such as joint pain, stiffness, chronic >6weeks, and worse with activity; without other causes of arthritis). X-ray is used for staging OA.
19. D.
20. E. We should generally avoid tramadol (opioid).
21. C.
22. D.
23. C, D, E. Enthesitis related arthritis is more common in boys, and the onset of age is often >10 years (early teens).
24. C. In all other types of JIA, uveitis is a serious yet asymptomatic complication that can potentially lead to blindness (essential that children get regular screening done). However, in enthesitis related JIA, uveitis is markedly painful (aka symptomatic) and red!

Week 12

Questions

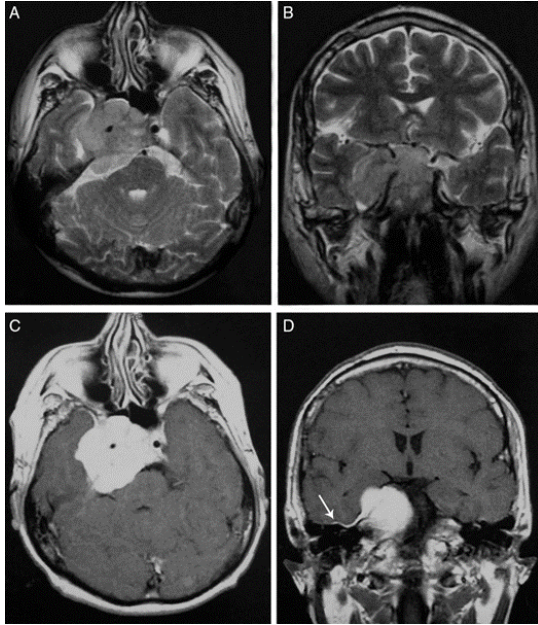
1. Which of the following statements regarding steroid use for rheumatologic disease is FALSE?
 - A. Intramuscular injections work better for patients who are non-compliant with oral steroids
 - B. Acute gout flare is an indication for high dose steroids
 - C. Side effects from chronic steroid use include weight gain, osteoporosis, and GI upset
 - D. Polymyalgia rheumatica is an indication for medium dose steroids
 - E. Recurrent UTIs are a contraindication for steroids
2. Which of the following statements regarding intra-articular injections is FALSE?
 - A. A red, hot, swollen joint should never receive an intra-articular steroid injection
 - B. Uncommon complications of intra-articular injections include tendon ruptures, post-injection flares, and infection
 - C. A diagnostic arthrocentesis should be sent for culture (+ Gram stain), cell count, and crystals
 - D. Hyaluronic acid injection can help relieve joint pain and is a form of therapeutic arthrocentesis
 - E. A joint aspirate containing a high number of neutrophils indicates a septic joint
3. Which of the following statements regarding medications for rheumatological conditions is FALSE?
 - A. Pregnancy is a contraindication for NSAID use
 - B. Opioids are uncommonly used in rheumatological disease
 - C. A patient with chronic kidney disease would be better off on a COXIB over an NSAID for their osteoarthritis
 - D. Acetaminophen is an anti-inflammatory that can be used safely up to 3g/day in divided doses in a healthy person
 - E. NSAIDs are indicated in rheumatoid arthritis but do not delay radiographic changes to bones and joints
4. Which of the following statements regarding DMARD and biologic use for rheumatologic conditions is FALSE?
 - A. Biologics are very expensive treatment but have the potential to drive some rheumatologic conditions into remission
 - B. Methotrexate requires baseline and monthly bloodwork to monitor for adverse effects like hepatotoxicity
 - C. Hydroxychloroquine requires annual eye exams due to the risk of retinal toxicity
 - D. Biologics act by targeting TNF and other inflammatory molecules
 - E. DMARDs are 3rd line medications, after treatment failure with steroids and NSAIDs

5. Which of the following statements regarding chronic neuropathic pain is FALSE?
- A. Individuals affected often adopt pain-avoidant behaviours
 - B. Pain is often perceived as being out of proportion and persistent despite resolution of the original injury
 - C. Decreased sleep and appetite are common somatic symptoms of chronic pain-associated depression
 - D. An individual with chronic neuropathic pain often describes their pain as sharp and stabbing in various locations
 - E. Descending pain modulation pathways like norepinephrine and serotonin have reduced activity
6. Jerry has had chronic long-standing back pain longer than you have been in practice. He had injured his back in a workplace incident over 20 years ago. You took him on from his now-retired previous family physician, who had always told Jerry to avoid bending his back to avoid aggravating the pain and injury further. Jerry considers himself a good patient and has always followed his doctor's orders exactly as instructed, but is frustrated as he feels he has limited a significant portion of his life due to his injury. Which of the following statements should you NOT say to Jerry?
- A. "I am truly sorry for all of the years you have suffered with your pain, and I am here to help"
 - B. "Exercising and stretching your back may help you reduce your pain"
 - C. "Your pain indicates that there is some ongoing tissue damage"
 - D. "How is your mood?"
 - E. "Your pain may actually be due to dysfunction in your brain's perception of pain"
7. Which of the following statements regarding pharmacologic modalities for managing chronic pain is NOT correct?
- A. Antidepressants often require a higher dose for addressing patient mood disturbances compared to the dose required for addressing pain
 - B. Long-term NSAID requires blood pressure monitoring
 - C. The major side effect associated with anticonvulsant use is sedation
 - D. Some components of the Opioid Risk Tool include family history of substance abuse, age 50-65, and psychological disease
 - E. Cannabis strains with higher CBD:THC ratios may play a beneficial role in pain management
8. Which of the following definitions regarding chronic pain is incorrect?
- A. In conversion disorder, there is incompatibility between the patient's symptoms and their neurological condition

- B. A diagnosis of somatic symptom disorder requires one or more somatic symptoms that are distressing or disruptive; thoughts, feelings, or behaviours related to the somatic symptoms; and chronicity of 6 months or greater
 - C. Medically unexplained symptoms are most common in neurology
 - D. If a patient presents themselves as injured but does not appear to have any obvious reason or reward for doing so, this is known as malingering
 - E. An adjustment disorder is described as emotional or behavioural symptoms that develop in response to a stressor, accompanied by out-of-proportion distress and/or impairment in function
9. Management of chronic pain requires a holistic approach that should involve collaboration with the patient, a multidisciplinary team, and is multimodal. Which of the following statements regarding management is FALSE?
- A. Common components of cognitive behavioural therapy include training in effective communication and use of distraction
 - B. It is best to prescribe pharmacologic agents in a step-wise approach
 - C. Patients may benefit from meditation, which includes breathing and visualization techniques
 - D. Cognitive behavioural therapy can be helpful individually as well as in groups
 - E. Physical management modalities, like transcutaneous nerve stimulation (TENS), tend to provide significant improvement to pain
10. What is your differential diagnosis for acute onset of aphasia? (Select all that apply.)
- A. Migraine
 - B. Post ictal Todd's paresis
 - C. Metabolic Derangement
 - D. Psychogenic
11. What body parts have the largest representations on the sensory homunculus?
- A. Arms and hands
 - B. Legs and feet
 - C. Face and hands
 - D. Trunk and genitals
12. What view of the brain is this?



- A. Sagittal
 - B. Coronal
 - C. Axial
 - D. Transverse
13. Lou had a stroke 1 week ago. You are assessing him at your family clinic and find that he has poor fluency, poor repetition, and poor naming, but can follow commands. Which area of his brain did his stroke most likely affect?
- A. R frontal lobe
 - B. L frontal lobe
 - C. R temporal lobe
 - D. L temporal lobe
 - E. R parietal lobe
14. Which of the following statements regarding cerebrum function is TRUE?
- A. The parietal lobe comprises the primary motor cortex and the ventral stream of the visual system
 - B. Occipital seizures often begin with auditory hallucinations and can be triggered by repetitive sounds
 - C. The anterior multimodal association area helps integrate information from past experience with information received from other areas of the brain to appropriately respond to stimuli
 - D. The function of the neurons in the postcentral gyrus is to send impulses to the spinal cord for motor activity
 - E. The face is mapped most medially on the sensory and motor homunculi
15. 53-year-old man presents with a history of daily stabbing headache associated with dizziness, progressive blurred vision, right ocular pain, ptosis and ophthalmoplegia. Upon examination, he is found to have a CN III palsy and ipsilateral Horner syndrome. Brain MRI identified a lesion homogeneously enhanced after gadolinium administration on T1-weighted images. Findings were consistent with a meningioma. What is at the top of your differential diagnosis?



- A. Diabetic third nerve palsy
 - B. Cavernous sinus syndrome
 - C. Third nerve compression via posterior communicating artery aneurysm
 - D. Midbrain stroke
16. Which of the following methods can be used to test the vestibulocochlear nerve (CN VIII)?
(Select all that apply)
- A. Weber test
 - B. Rinne test
 - C. Nystagmus test
 - D. Romberg test
17. 29-year-old male presents with a 1-year history of headaches, weight loss, decreased muscle strength, stamina, decreased libido and morning erections. He shaves every second day. Investigations showed decreased testosterone(nmol/L) but elevated prolactin (1257ug/l). MRI showed a pituitary mass. What is at the top of your differential diagnosis, and what cranial nerve should you test?

- C. Kallmann syndrome and Olfactory Nerve (CN I)
 - D. Macroprolactinoma and Optic Nerve (CN II)
19. What supplies sensation over the ear and the skin in the temporal region?
- A. Frontal Nerve
 - B. Infraorbital Nerve
 - C. Mental Nerve
 - D. Great Auricular Nerve
 - E. Auriculotemporal Nerve
20. What supplies sensation over the angle of the mandible?
- A. Frontal Nerve
 - B. Infraorbital Nerve
 - C. Mental Nerve
 - D. Great Auricular Nerve
 - E. Auriculotemporal Nerve
21. Which of the following pairings are **incorrect**?
- A. Frontalis – raises eyebrows
 - B. Zygomaticus major – pulls mouth
 - C. Orbicularis oris – closes eye
 - D. Platysma – helps to depress mandible
 - E. Occipitalis – moves scalp
22. What are the muscles of mastication?
- A. Masseter and temporalis
 - B. Orbicularis oris and orbicularis oculi
 - C. Zygomaticus major and zygomaticus minor
 - D. Buccinator and platysma
23. Which of the following pairings are **correct**? (Select all that apply.)
- A. Temporal – Frontalis
 - B. Zygomatic – Zygomaticus Major
 - C. Buccal – Buccinator
 - D. Mandibular – Orbicularis Oris
 - E. Cervical - Platysma
24. 69-year-old woman presents with right-sided hemiparesis, right-sided hemisensory loss, right-sided homonymous hemianopia, and aphasia. Past medical history is significant for hypertension, hypercholesterolemia, and diabetes mellitus. What vessel has been occluded?
- A. Right anterior cerebral artery (ACA)
 - B. Right middle cerebral artery (MCA)
 - C. Left middle cerebral artery (MCA)
 - D. Left posterior cerebral artery (PCA)
25. What would you be concerned about if the interventricular foramen become blocked?
- A. Microcephaly
 - B. Meningitis
 - C. Hydrocephalus

D. Craniosynostosis

26. Match the clinical condition with neuronal structure affected.

1. Parkinson's disease	A) Striatum
2. Huntington's disease	B) Nucleus Accumbens
3. Schizophrenia	C) Substantia Nigra

27. What would you expect to see on post-mortem examination of Parkinson's disease brains?

- A. Significant shrinkage of the brain
- B. Significant enlargement of the brain
- C. Loss of neuromelanin pigmentation in the substantia nigra
- D. Excess neuromelanin pigmentation in the substantia nigra

28. 45-year-old man has been experiencing an increase in accidents at work over the last couple of months, such as accidentally cutting his fingers with his knife and dropping plates. In the emergency room, you (the physician) notice that he was stumbling often and had difficulty with his upper limb coordination. He told you that he has had a difficult time at work using his knife recently and mentioned that "it must be due to being tired". You also noted that the man had a short temper and got angry with some of the other patients in the waiting room. MRI of the brain showed ventricular enlargement and degeneration of cortical layers. Genetic testing showed over 69 CAG repeats on the short arm of chromosome 4. What is at the top of your differential diagnosis?

- A. Huntington's disease
- B. Parkinson's disease
- C. Schizophrenia
- D. Cerebellar ataxia due to alcoholism

29. Which of the following is not a positive symptom of schizophrenia? (Select all that apply.)

- A. Hallucinations
- B. Delusions
- C. Anhedonia
- D. Alogia

Answers

1. **B.** An acute gout flare is an indication for medium dose steroids (25-30 mg daily x 3 weeks → taper by 5 mg every 3 days). High dose steroid indications include ANCA-associated vasculitis, lupus nephritis, inflammatory myositis, and giant cell arteritis.

2. **A.** A red, hot, swollen joint could benefit from an intra-articular steroid injection if the patient is experiencing an acute gout flare or pseudogout, etc. Not every red, hot, swollen joint is a septic joint, but this should be ruled out by performing a diagnostic arthrocentesis.
3. **D.** Acetaminophen is not an anti-inflammatory medication. It is an analgesic. It can be used safely up to 3g/day in divided doses in a healthy person but should be reduced in liver disease.
4. **E.** DMARDs are best used as early and aggressively as possible to prevent early joint damage. Most also take many weeks for their onset of action, so they ideally should be initiated within the first 3 months of symptom onset.
5. **D.** Chronic neuropathic pain can present similarly to nociceptive pain, but is generally described as having a burning, electrical, or pins and needles numbness quality.
6. **C.** It is likely that Jerry's back injury healed long ago. His pain does not indicate ongoing tissue damage. It is instead likely due to dysfunction in his pain signal modulation. Exercising and flexibility training may improve his pain. Acknowledging his pain and what he has gone through is a decent thing to do. Many with chronic pain suffer from depression and anxiety and these should not be missed.
7. **D.** The age group at greatest risk for opioid addiction is 16-45. Other risk factors are male, personal or family history of substance abuse, history of preadolescent sexual abuse, and psychological disease.
8. **D.** Malingering is when a patient is consciously acting to falsify their symptoms for external gain. The situation in D is more descriptive of a factitious disorder, where the patient is conscious of their behaviour but has no reward for doing so.
9. **E.** Although they may be helpful for some patients to some degree, physical management modalities tend to only provide modest improvements to chronic pain.
10. **A,B,C,D;** Differential diagnosis for acute onset of aphasia includes ischemic stroke, intracranial hemorrhage, migraine, post ictal Todd's paresis, metabolic derangement in the setting of underlying structural lesion and psychogenic.
11. **C**
12. **B**
13. **B.** The findings of poor fluency, repetition, and naming, with preserved command following, would be seen in Broca's aphasia. Broca's area is found in the frontal lobe. It is most commonly found in the patient's dominant hemisphere, which most often is their L hemisphere.
14. **C.** The description of the anterior multimodal association area is true. The parietal lobe comprises the somatosensory cortex and the dorsal stream of the visual system. Occipital seizures often begin with visual hallucinations and can be triggered by flashing lights or repeating patterns. The function of the neurons in the precentral gyrus is to send impulses to the spinal cord for motor activity, not the postcentral gyrus; the postcentral gyrus contains the primary somatosensory cortex. The face is mapped more laterally on the sensory and motor homunculi.
15. **B.** Cavernous sinus syndrome (CSS) is a condition caused by any pathology involving the cavernous sinus which may present as a combination of unilateral ophthalmoplegia (cranial nerve (CN) III, IV, VI), autonomic dysfunction (Horner syndrome) or sensory CN V1- CN V2 loss. The common clinical findings of CSS are as follows: Total/partial ophthalmoplegia: CN III palsy,

CN IV palsy, CN VI palsy, Facial sensory loss: CN V1 loss and CN V2 loss, Horner syndrome: loss of sympathetic tone due to damage of sympathetic plexus, and Proptosis and chemosis: due to increased pressure within cavernous sinus. A CSS is caused by any pathology or lesion present within the cavernous sinus that disrupts the function of other anatomical structures. The most common cause of CSS is mass effect from tumor. Other common causes of CSS include trauma and self-limited inflammatory disease. Less common causes are vascular etiologies and infections.

16. **A,B,C,D.**
17. **D.** Macroprolactinoma can cause compression of the optic chiasm which disrupts the nasal fibers of the optic nerve leading to a bitemporal visual field defect. Refer to “Clinical Presentation and Evaluation of Pituitary Gland Disease” module, slide 7-9.
18. **C.** Kallmann syndrome (KS) is a genetic disorder that prevents a person from starting or fully completing puberty. Kallmann syndrome is classified under hypogonadotropic hypogonadism. To distinguish it from other forms of hypogonadotropic hypogonadism, Kallmann syndrome has the additional symptom of a total lack of sense of smell (anosmia) or a reduced sense of smell. Refer to “Disorders of Puberty”, slide 31.
19. **E**
20. **D**
21. **C.** Orbicularis oris closes the mouth
22. **A**
23. **A,B,C,D,E**
24. **C.** Complete MCA strokes will primarily cause a visual field loss or homonymous hemianopia to the opposite side because they affect the optic radiations projecting to the occipital cortex. MCA strokes on the left side can cause an aphasia, right sided hemiparesis, and right sided hemisensory loss.
25. **C.** If an interventricular foramen became blocked, cerebrospinal fluid could not flow from the lateral ventricles into the third ventricle. Cerebrospinal fluid would continue to form within that ventricle, so the blocked ventricle would swell with fluid - a condition known as hydrocephalus.
26. **1C, 3B, 2A**
27. **C.** Post-mortem examination of Parkinson’s disease brain at a macroscopic level shows no brain shrinkage; there is not significant difference between Parkinson’s disease brains and age-matched controls in the number of neurons. In the Parkinson’s disease brain, there is significant and specific degeneration of the dopaminergic neuron of the substantia nigra.
28. **A.** MRI findings and genetic testing are consistent with Huntington’s disease.
29. **C,D**

Week 13

Questions

1. A 51 year-old woman traveling across Canada can see the beautiful blue sky with white clouds but is unable to focus on her face in the mirror. Her son notices a distinct droopy eyelid as well as an inferolaterally directed right eye and takes her to a walk-in clinic. What is this woman most likely experiencing?
 - a. Bell's Palsy
 - b. Third Nerve Palsy
 - c. Fourth Nerve Palsy
 - d. Sixth Nerve Palsy
 - e. Horner Syndrome
2. A 7 year old boy is only able to produce tears from his left eye following a tooth abscess from a long-standing infection. The infection spreads along the nerves of the ANS. Which of the following nerve pathways is possibly affected to result in his symptoms?
 - a. CN III
 - b. CN IV
 - c. CN V₁
 - d. CN V₂
 - e. CN V₃
3. You are performing a test for visual fields. Your patient correctly counts a total of one when you hold up you hold up one finger on the left side. However, when you simultaneously hold up one finger on both sides, she counts a total of one as well. Where is she most likely to have a lesion?
 - a. Supplementary motor area
 - b. Prefrontal cortex
 - c. Posterior parietal cortex
4. Axon tracts responsible for finger movements in playing the piano decussate at the:
 - a. Cerebellum
 - b. Brainstem
 - c. C4 vertebral level
5. Billie Eilish constantly wriggles her ears back and forth, raises her eyebrows, clicks her jaws, and flexes her arms as a result of a condition she was diagnosed with at age 11: Tourette's Syndrome. Which structure is most likely affected in this disorder?
 - a. Vestibulo-cerebellum
 - b. Spinocerebellum
 - c. Cerebrocerebellum
 - d. Basal Ganglia
6. Compare and contrast Huntington's Disease vs. Parkinson's Disease.

7. A 92 year old gentleman comes in with a complaint of loss of movement in his hands but not shoulders. He has recently experienced a fall. Which lesions of the spinal cord are most likely in this case?
 - a. Anterior cord injury
 - b. Posterior cord injury
 - c. Central cord injury
 - d. Brown Sequard
8. Match the the correct location of decussation to each of the following tracts: corticospinal, spinothalamic, medial lemniscus
 - a. Medulla
 - b. Spinal cord entry
9. A 48 year old female complains of double vision. She is currently on ramipril, hydrochlorothiazide, metformin, and acetazolamide. Cranial nerve exam reveals an intorted left eye, but pupil sizes are normal. What is the most likely cause of her diplopia?
 - a. Unruptured berry aneurysm
 - b. Ruptured berry aneurysm
 - c. Diabetes complication
 - d. Congenital
10. A patient presents with bitemporal hemianopia. What is the most probable lesion?
 - a. Pituitary tumor
 - b. MCA stroke
 - c. PCA stroke
 - d. Right optic nerve injury
11. A patient has a lesion in the left occipital lobe. Which of the following may she present with?
 - a. Loss of blink reflex in the right eye
 - b. Loss of blink reflex in the left eye
 - c. Neglect of items on the right side
 - d. Neglect of items on the left side
12. A 73 year old woman wakes up unable to find words in her speech. Where is the lesion most likely localized?
 - a. Cortical
 - b. Subcortical
 - c. Brainstem
13. If the optic nerve is damaged, which affected artery can lead to blindness?
 - a. Middle meningeal artery
 - b. Central retinal artery
 - c. Supratrochlear artery
 - d. Dorsal nasal artery
 - e. Posterior ciliary artery
14. Which of the following structures will be affected by a cavernous sinus thrombus (multiple correct answers)?
 - a. Optic chiasm

- b. CN IV
 - c. CN VIII
 - d. CN V3
 - e. CN V2
 - f. Internal carotid artery
15. Which of the following states are true (multiple correct answers)?
- a. the major inhibitory neurotransmitter in the brain is glutamate
 - b. Upon physical contact with an object, the neurons convey the length of contact through the frequency of action potential fired.
 - c. Acetylcholine is the excitatory neurotransmitter at the neuromuscular junction.
 - d. Ionotropic receptors transmit signals through a second –messenger system
 - e. The GABA receptor is a metabotropic receptor.
16. What is the difference Broca's aphasia and Wernicke's aphasia:
- a. Broca's aphasia involves injury to the frontal lobe, therefore patients have disrupted comprehension. Wernicke's aphasia involves injury to the parietal lobe, therefore patients have disrupted language expression.
 - b. Broca's aphasia involves injury to the temporal lobe, therefore patients have disrupted comprehension. Wernicke's aphasia involves injury to the frontal lobe, therefore patients have disrupted language expression.
 - c. Broca's aphasia involves injury to the frontal lobe, therefore patients have disrupted language expression. Wernicke's aphasia involves injury to the temporal lobe, therefore patients have disrupted comprehension.
 - d. Broca's aphasia involves injury to the parietal lobe, therefore patients have disrupted comprehension. Wernicke's aphasia involves injury to the temporal lobe, therefore patients have disrupted language expression.
17. Which component of the eye is responsible for accommodation
- a. The lens and ciliary muscles
 - b. The iris and pupil
 - c. The ciliary muscle and pupil
 - d. The pupil and cornea
 - e. The posterior chamber and fovea
18. Which of the following eye pathology is caused by increased intra-ocular pressure compressing axons of retinal ganglion cell
- a. cataract
 - b. Horner syndrome
 - c. Macular degeneration
 - d. Glaucoma
 - e. Myopia
19. Which of the following is involved in the formation of memory (multiple correct answers)
- a. Hippocampus
 - b. Medial frontal cortex
 - c. Ventromedial prefrontal cortex

- d. Amygdala
- 20. What type of memory impairments are involved in medial frontal syndrome
 - a. Anterograde and retrograde memory
 - b. Utilization behavior
 - c. Abnormal working memory
 - d. Deficient spatial memory
- 21. Which of the following will appear white/bright on a T2-weighted MRI
 - a. CSF
 - b. White matter
 - c. Grey matter
 - d. Skull
 - e. Optic tract
- 22. Which of the following would be the most effective in diagnosing an aortic dissection
 - a. MRI
 - b. CT with contrast
 - c. CT without contrast
 - d. Ultrasound
 - e. Radiography
- 23. What of the following is not involved in the ascending pathway of the cerebellum
 - a. Thalamus
 - b. Pons
 - c. Superior cerebral peduncle
 - d. Motor cortex
- 24. Which of the following is a presentation of cerebral ataxia
 - a. Tendency to fall bilaterally
 - b. Tendency to fall ipsilaterally
 - c. Tendency to fall contralaterally
 - d. Flexor rigidity
- 25. Which pathway is responsible for pain and temperature sensation and where does this path decussate
 - a. Spinothalamic pathway, which decussates at the level of the nerve root entering the spinal cord
 - b. Dorsal column pathway, which decussates at the level of the nerve root entering the spinal cord
 - c. Spinothalamic pathway, which decussates at the medulla
 - d. Dorsal column pathway, which decussates at the medulla
- 26. What is the large touch receptor responsible for deep touch and arranged in layered concentric capsules
 - a. Pacinian corpuscle
 - b. Ruffini cylinder
 - c. Meissner's corpuscle
 - d. Merkel's receptors

27. You are called in to assess a newborn without a skull or the front portion of their brain. Which of the following diagnoses is most likely?
- Anencephaly
 - Lissencephaly
 - Microlissencephaly
 - Polymicrogyria
28. In what trimester does the neural tube close?
- First
 - Second
 - Third

Answers

- B. CN III Palsy presents with pupil dilation, intorsion of the eye (directed down and out), complete ptosis, and lack of accommodation.
- D. Post-synaptic parasympathetics for lacrimal gland innervation travels with V₂.
- C. The posterior parietal cortex is responsible for spatial attention. Lesions lead to neglect.
- B. Lateral corticospinal tracts responsible for fine finger movements decussate between the medulla and spinal cord.
- D. The basal ganglia is responsible for releasing appropriate movements and suppressing inappropriate ones.
- Huntington's is a hyperkinetic disorder caused by too little inhibition on the thalamus through the basal ganglia. Parkinson's is a hypokinetic disorder caused by a loss of dopaminergic cells in the substantia nigra pars compacta.
- C. Central cord injury corresponds with a suspended sensory loss of pain and temperature, as well as a loss of motor function in the hands but not the proximal arm.
- Medulla: Corticospinal, Medial lemniscus
Spinal cord entry: Spinothalamic
- C. Intorted eye is a 3rd nerve palsy, with parasympathetics spared. Diabetes affects the microvasculature, which can spare the parasympathetic innervation for pupil size.
- A. Pituitary tumors tend to compress the optic chiasm, affecting the temporal visual fields which are crossing there.
- C. Blink reflex is still intact in patients with visual neglect. The occipital lobe sees items on the contralateral side.
- A. Aphasia is a cortical sign.
- B. the central retinal artery and vein runs within the sheath of the optic nerve.
- A, B, E, F.
- C. Glutamate is the major excitatory neurotransmitter in the brain. Axons convey the length of physical contact with an object through the duration of action potential firing. Metabotropic receptors transmit signals through a second-messenger system, not ionotropic receptors, which opens ion channels open binding of neurotransmitters. The GABA receptor is a ionotropic receptor.

16. C
17. A.
18. D
19. A, B, C, D, E
20. A. Utilization behavior can be found in dorsolateral syndrome, which involves connections between the dorsal lateral prefrontal cortex with the thalamus, hypothalamus, amygdala, and hippocampus. Abnormal working memory can be a result of damaged dorsolateral prefrontal cortex.
21. A. fluid is white on T2 – weighted MRI, which could indicate CSF, edema, etc.
22. D. Ultrasound has a very high temporal resolution, which is useful in visualizing the dissection flap and blood flow.
23. B. The pons is involved in the descending pathway, but not ascending pathway of the cerebellum.
24. B. The cerebellum controls the motor function of the ipsilateral body.
25. A.
26. A. Pacinian corpuscle and Ruffini cylinder are deep touch receptors, but Ruffini cylinder receptors consists of many branched fibres in a cylindrical arrangement. Meissner's corpuscle and Merkel's receptors are shallow touch receptors.
27. A.
28. A. The neural tube closes at D21-24, roughly Week 6 of development. Therefore, early folic acid supplementation is essential.

Week 14

Questions

1. Which of the following statements is **TRUE** regarding UMN vs LMN lesions?
 - A) LMN lesions may present with reduced or absent reflexes.
 - B) LMN lesions may present with hyperreflexia and spasticity.
 - C) UMN lesions can present with muscle atrophy while LMN lesions can present with normal bulk or atrophy.
 - D) UMN lesions can present with weak arm extensors and leg flexors.
 - E) UMN lesions can localize to the brain, brainstem, spinal cord, or anterior horn cells.

2. Based on the presentation, identify the cerebral artery territory *most likely* impacted. Answers may be used more than once.
 - A) ACA
 - B) MCA
 - C) PCA
 - 1) ____ can follow commands but produces nonsensical speech. Severe arm weakness, facial droop.
 - 2) ____ moderate leg weakness, mild arm weakness.
 - 3) ____ Paralysis of left arm, leg and lower face. Complete sensory loss in left arm, leg and face.
 - 4) ____ dizziness, diplopia, contralateral homonymous hemianopsia.
 - 5) ____ neglect of left side, left asomatognosia, left hemianesthesia, left face and arm weakness.

3. Match the following localizations to the patient presentations:
 - A) cortex
 - B) subcortical
 - C) brainstem
 - D) spinal cord
 - E) peripheral nerve
 - F) neuromuscular junction
 - G) muscle
 - 1) ____ 80 y/o man with numb feet and reduced pin-prick sensation over medial/lateral/posterior leg. Bilateral dorsiflexion and plantarflexion were 4's, remainder of power is full. Absent ankle reflexes, other reflexes normal.hm
 - 2) ____ 55 y/o man with moderate weakness of bilateral deltoids, triceps, hip flexors and hip abductors. Normal reflexes, tone, and sensation.

- 3) ____ 29 y/o woman with reduced sensation in left face, left arm, and left leg.
 - 4) ____ 32 y/o woman with ptosis that gets worse through the day. Normal reflexes and sensation.
 - 5) ____ 40 y/o woman with progressive balance difficulties. 4/5 hip flexors, 4/5 knee flexors, and 4/5 dorsiflexors, all bilaterally; other lower and upper extremity power testing was normal. Hyperreflexia at knees and ankles. Reduced pinprick sensation diffusely over leg extending up to umbilicus.
 - 6) ____ 45 y/o man with vertigo and headache. Reduced sensation noted in right face, left arm extensors, and left hip flexors.
 - 7) ____ 77 y/o woman unable to see objects in left visual field. Loss of sensation in left face and left arm.
4. A 25 year-old man is brought to the ER following a high-speed motor vehicle accident. What should you **NOT** include in your immediate management and investigations?
- A) C-spine precaution and intubation
 - B) Large bore IVs
 - C) Bloodwork
 - D) STAT CT
 - E) MRI
5. Which statement is **FALSE** regarding severe traumatic brain injuries?
- A) Subdural hematomas show up as “crescent-shaped” on CT, while intracerebral hemorrhages show mixed hyper- and hypodense areas on CT.
 - B) Basilar skull fractures may present with raccoon eyes, Battle sign, and CSF rhinorrhea.
 - C) Subdural hematomas result from an acceleration/deceleration injury that tears the bridging vessels.
 - D) Subarachnoid hemorrhages often co-exist with diffuse axonal injury.
 - E) Cerebral contusions can result in bruising of the brain beneath and/or opposite the impact site.
6. Which statement is **TRUE** in the diagnosis and management of concussion?
- A) The Concussion Recognition Tool (CRT5) is used as a diagnostic aid by a physician in clinic while the Sport Concussion Assessment Tool (SCAT5) is used as a diagnostic aid by a coach on the field.
 - B) If a player suffers an impact to the head during a game but rests until they are symptom-free, they may return to play with added precautions.
 - C) Extended rest for 2 weeks following a concussion is recommended.
 - D) Sending a player to the ER following a hit to the head is indicated if they show deteriorating mental status, worsening symptoms, or a GCS < 15.
 - E) To diagnose a concussion, a minimum of 3 concussion symptoms and mechanism of injury involving head impact (either direct or indirect) is required.

7. Which statement is **TRUE** regarding multiple sclerosis (MS)?
- A) Multiple sclerosis is most commonly found in 40-60 year old females of Caucasian/Northern European descent.
 - B) Primary progressive MS is often seen in young men with genetic risk factors.
 - C) Investigations for diagnosing MS include MRI and lumbar puncture.
 - D) Vitamin D deficiency is not a risk factor for MS.
 - E) Management of MS includes disease modifying therapies, treatment of symptoms, and surgical consult for myelin grafts.
8. Which is NOT a treatment option used for neuropathic pain?
- A) Amitriptyline
 - B) NSAIDs
 - C) Gabapentin
 - D) Capsaicin
 - E) Cannabinoids
9. Which of the following best describes the typical progression of symptoms for an epidural hematoma?
- a. Acute post-traumatic coma
 - b. A period of alertness followed by a rapid descent into unconsciousness
 - c. Headaches, dizziness, alteration of alertness
 - d. Nausea, vertigo, and rapid descent into unconsciousness
10. What is the appropriate management for an acute hydrocephalus?
- a. External ventricular drainage
 - b. Ventriculoperitoneal shunt
 - c. Mannitol
 - d. Decadron
11. A patient presents with a month-long progressive weakness running up from her foot to her knees to the point where she is having trouble walking for long periods of time. You find reduced Achilles and patellar reflexes on physical exam. What is at the top of your differential?
- a. Myasthenia gravis
 - b. Multiple sclerosis
 - c. Guillain Barre syndrome
 - d. Radiculopathy
12. What is the pathophysiology of distal dying back neuropathy?
- a. Degeneration of the distal portion of the axon, progressing proximally
 - b. Degeneration of the distal portion of the axon, progressing distally
 - c. Transient dysfunction of the ion channels
 - d. Degeneration of the neuronal cell body

13. A patient was involved in a MVA and fell into unconsciousness at the scene. Upon examination, you find that they are not able to open their eyes, have no verbal communication, but are able to flex their arm towards a painful stimulus. What is the GCS score for this patient?
- 4
 - 5
 - 6
 - 7
14. You are evaluating a patient during your morning rounds. They are unconscious, and because you haven't yet reviewed their chart, you are unaware of the cause. However, you observe that they are lying with straight legs but bent elbows and wrists. You can thus surmise that the lesion is likely:
- Within the cerebral cortex
 - Within the midbrain
 - Within the medulla
 - Within the spinal cord
15. Which of the following is NOT diagnostic of epilepsy?
- At least 2 unprovoked seizures within 24hrs
 - One unprovoked seizure and an abnormal investigation that indicates high risk of recurrence
 - Diagnosis of an epileptic syndrome
 - One unprovoked seizure and a probability of further seizures occurring over the next 10 years
16. What is the difference between epileptic and non-epileptic seizures?
- Epileptic seizures are due to an epilepsy disorder while non-epileptic seizures are not
 - Epileptic seizures have a motor component while non-epileptic seizures do not
 - Epileptic seizures are recurring while non-epileptic seizures are not
 - Epileptic seizures have an abnormal EEG finding while non-epileptic seizures do not
17. Mr. Diaz is a 72 year old gentleman who was found collapsed on his kitchen floor by his wife. He was found to have a droopy face on his left side, and weakness in his left arm and leg. He was rushed to the emergency room on a stroke protocol. He was found to have a BP of 140/80, normal glucose, and no current medications. Does he currently show any contraindications to tPA?
- Yes, due to his advanced age
 - Yes, due to his hypertension
 - Yes, due to the time from onset
 - No

18. What is the first and primary imaging modality for patients with suspected strokes?
- CT
 - CTA
 - MRI
 - MRA

Answers

- D. UMN lesions follow a pyramidal pattern of weakness (patients may have their arm flexed to chest and walk with a straight leg (because of stronger arm flexors and leg extensors). Incorrect answers: A – LMN lesions can also present with normal reflexes. B – both are characteristic of UMN lesions. C – UMN lesions present with normal bulk, LMN lesions present with atrophy. E – anterior horn cells are a localization of LMN lesions.
- 1) B – left MCA. 2) A – ACA (clue is leg > arm weakness). 3) B – MCA, likely internal capsule. 4) C 5) B – right MCA.
- 1) E peripheral nerve – sensory stocking distribution, distal weakness in foot. 2) G muscle – symmetric, proximal weakness in the shoulder and hip girdle with no sensory involvement. 3) B subcortical sign of pure sensory hemibody affecting face, arm, and leg equally. 4) F NMJ – fatiguing weakness in ocular muscles. 5) D – spinal cord signs of pyramidal weakness with weak leg flexors, hyperreflexia suggesting UMN lesion, and sensory level at T10. 6) C brainstem – vertigo + another symptom, crossed sensory symptoms, pyramidal weakness. 7) A cortical sign of contralateral homonymous hemianopsia and contralateral sensory loss.
- E. MRI is usually ordered after an emergent/screening CT.
- A. Cerebral contusions show mixed density areas on CT.
- D. indications for urgent referral to ER include GCS < 15, deteriorating mental status, potential spinal injury, and progressive/worsening symptoms or new neurologic signs. A) CRT is used on the field and SCAT5 is used in clinic. B) if concussion is suspected, players are NOT allowed to return to play until assessed by a medical professional. C) extended rest is NOT recommended at all. E) only 1 concussion symptom is needed for diagnosis.
- C. A) 20-40 year old females. B) older men.
- B. NSAIDs are used for treating nociceptive pain.
- B. Lucid period followed by rapid deterioration is characteristic of epidural hematoma. Intracranial pressure progressively builds as blood accumulates within the epidural space.
- A. Ventriculoperitoneal shunt is for chronic hydrocephalus, the other two are medical modalities for decompression only (vs. drainage).
- C. The presentation is consistent with a LMN disease, more specifically a polyneuropathy. The progressive ascending weakness is characteristic of GBS.
- A. Distal dying back often due to impaired transport down the axon, resulting in degeneration starting from the distal portion of the axon. B best describes Wallerian degeneration, C best describes functional block, and D best describes neuronopathy.

13. B. No eye opening or verbalization is a 1 for each category. Flexion towards painful stimuli is an abnormal flexion response, which is a 3 for the motor response category.
14. B. The patient is presenting with a decorticate response or flexor posturing, which is indicative of a lesion above the red nucleus (within the midbrain or above).
15. A. Multiple seizures occurring within 24hrs could be indicative of a common unknown provoking factor and would not be diagnostic. At least 2 unprovoked seizures more than 24hrs apart would be a diagnostic criteria of epilepsy.
16. D. This is the definition of epileptic and non-epileptic seizures.
17. C. Because he experienced an unwitnessed onset, we cannot know for sure how long it has been since onset. TPA would thus be contraindicated in case it has been too long to be effective.
18. A. Start with CT, and then can do CTA to assess for large vessel occlusion. MRIs not routinely done in a stroke protocol.

Week 15

Questions

1. Part 1 of 2: Tom Paris presents with complaints of foot drop and difficulty swallowing. On history it is discovered that the weakness has been progressively getting worse over a number of months, and on physical examination you discover signs of atrophy, fasciculations, hyperreflexia, and spasticity. There is no evidence of sensory, bladder, or coordination issues. These findings are consistent in both the cervical and the lumbosacral regions. You order an MRI and are able to rule out any structural causes for these symptoms. What is your most likely diagnosis?
 - a. MS
 - b. ALS
 - c. Radiculopathy
 - d. Spondylosis
 - e. Dermatomyositis
2. Part 2 of 2: What pharmacological treatments are available for the condition Tom Paris is suffering from? Select all that apply.
 - a. Corticosteroids
 - b. Riluzole
 - c. Plasmapheresis
 - d. Edaravone
 - e. There are no pharmacological treatments available.
3. Where is weakness associated with ALS localized to?
 - a. Brain
 - b. Spinal Cord
 - c. Motor Neuron/AHC
 - d. Root
 - e. Plexus
 - f. Nerve
 - g. NMJ
 - h. Muscle
4. A 3 month old female presents with low tone and mildly dysmorphic features. On examination she has appropriate strength for her age, brisk reflexes, and fisted hands, and no signs of muscle atrophy. Where does this lesion localize to?
 - a. Upper motor neuron
 - b. Lower motor neuron
 - c. Both upper and lower motor neuron
 - d. Connective tissue disorder
5. What is the primary method used to diagnose cerebral palsy?
 - a. MRI
 - b. CT scan

- c. Physical examination
 - d. Genetic testing
6. What is the most common cause of cerebral palsy?
- a. Lack of oxygen to the brain
 - b. Infections during pregnancy
 - c. Traumatic brain injury
 - d. Genetic factors
7. What is the most common type of cerebral palsy?
- a. Spastic
 - b. Ataxic
 - c. Athetoid
 - d. Mixed
8. What are the potential complications of a mitochondrial disorder?
- a. Seizures
 - b. Difficulty swallowing
 - c. Heart problems
 - d. All of the above
9. You suspect your patient has a mitochondrial disorder. You decide to run several test to confirm your suspicions. Which of the following findings would you expect, that make a mitochondrial disorder more likely? Select all that apply.
- a. Increased blood lactate
 - b. Decreased CSF lactate
 - c. Mitochondrial DNA and nuclear mitochondrial gene panels return positive
 - d. MRI/MRS showing basal ganglia changes and increased muscle atrophy
10. A patient presents with symptoms of a neurodegenerative disorder, including movement problems, cognitive decline, and behavioral changes. They have a family history of similar symptoms, and genetic testing reveals a mutation in the HTT gene. What is the most likely diagnosis?
- a. Alzheimer's disease
 - b. Parkinson's disease
 - c. Huntington's disease
 - d. Amyotrophic lateral sclerosis (ALS)
11. How many repeats is considered diagnostic for Huntington disease in the HTT gene?
- a. >30
 - b. >40
 - c. >50
 - d. >60
12. Which of the following are early symptoms associated with increased ICP? Select all that apply.
- a. Decreased LOC
 - b. Headache
 - c. Nausea and vomiting
 - d. Papilledema

- e. Cushing response
 - f. Herniation
13. Which type of the following gliomas is typically considered WHO Grade III?
- a. Glioblastoma multiforme
 - b. Low grade glioma
 - c. Pilocytic astrocytoma
 - d. Anaplastic Glioma
14. A 55-year-old woman comes to the physician because of a 6-month history of worsening involuntary movement of the left hand. She also reports that it takes her longer than usual to get up from a chair. Her symptoms are worse when she feels stressed at work. She has no history of serious illness and takes no medications. Neurological examination shows difficulty initiating movement and a tremor in the left hand at rest. The tremor decreases when the patient is asked to draw a circle. Which of the following is the most appropriate pharmacotherapy?
- a. Methimazole
 - b. Pramipexole
 - c. Donepezil
 - d. Clonazepam
15. What are the components of the T.R.A.P. acronym for the typical symptoms of Parkinson's
- a. Tumor, Rigidity, Aphasia, Parasthesia
 - b. Tremor, Rigidity, Akinesia, Paresis
 - c. Tremor, Rigidity, Akinesia, Postural Instability
 - d. Tumor, Rigidity, Akinesia, Polyarthralgia
16. A previously healthy 3 month-old infant is brought to the emergency department 6 hours after the onset of fever and persistent irritability. He had been well until 2 days ago, when he started feeding poorly and sleeping more than usual. He appears lethargic and irritable when roused for examination. His temperature is 39.0°C (102°F). He cries when he is picked up and when his neck is flexed. The remainder of the physical and neurological examinations show no other abnormalities. A gram stain of his lumbar puncture CSF showed **Gram negative coccobacilli**. What organism is responsible?
- a. E. coli
 - b. N. meningitidis
 - c. S. Pneumoniae
 - d. H. influenzae type B
17. In an immunocompetent ADULT presenting with acute bacterial meningitis which IV medication regime is most appropriate?
- a. Penicillin, Ancef, and Hydrocortisone
 - b. Flagyl, Rifampin, Dexamethasone

- c. Pip-tazo, Vancomycin, Ibuprofen
- d. Ceftriaxone, Vancomycin, Dexamethasone

18. When neuronal regeneration goes awry and the axon haphazardly sprouts synkinesis can happen.

What is an example of synkinesis?

- a. Innervation to a new target
- b. Innervation to the same target
- c. Fusion of two nerves
- d. New growth with faster conduction

19. A 42-year-old woman comes to the physician because of increasing fatigue and difficulty concentrating at work for the past 2 months. She has hypertension and a 22-year history of Crohn disease. She has been hospitalized and treated for acute exacerbations involving strictures multiple times in the past. She has not had significant gastrointestinal symptoms in over a year. Current medications include mesalamine, hydrochlorothiazide, and bisoprolol. Her temperature is 37.2°C, pulse is 72/min, and blood pressure is 140/90 mm Hg. Examination shows pale conjunctivae and a soft abdomen, she has a positive Romberg test, and has been finding herself almost losing balance in the shower and feels unsteady on the stairs, she has 3+ reflexes in Biceps, knee, and Achilles. Rectal examination is unremarkable, and she has no urinary symptoms. Laboratory studies show: Hgb 94, MCV 112, Iron studies normal, TSH normal

Further physical evaluation of this patient is most likely to show which of the following findings?

- a. Tarry stools
- b. Impaired vision in her right eye
- c. Unexplained weight loss
- d. Decreased vibration sensation

20. A 58-year-old woman is brought to the emergency department because of a 2-day history of increasing chest pain and shortness of breath. She has had a productive cough with foul-smelling sputum for 1 week. Seven months ago, the patient had an ischemic stroke. She has smoked one pack of cigarettes daily for 40 years. She has a 20-year history of alcohol use disorder but has not consumed any alcohol in the past 4 years. Her only medication is omeprazole. She appears to be in respiratory distress and speaks incoherently. Her temperature is 39.3°C, pulse is 123/min, respirations are 33/min, and blood pressure is 155/94 mm Hg. Auscultation of the lung shows rales and decreased breath sounds over the right upper lung field. A chest x-ray shows infiltrates in the right posterior upper lobe. Which of the following is the strongest predisposing factor for this patient's aspiration pneumonia?

- a. Smoking history
- b. Alcohol misuse history
- c. PPI use

d. History of ischemic stroke

Answers

1. B - ALS can be diagnosed when you have a consistent history of relatively isolated motor symptoms which are progressive, exam demonstrates a combination of LMN and UMN signs in at least two regions (without prominent extra-motor involvement), and other investigations have ruled out other explanations for the symptoms.
2. B and D. Corticosteroids and plasmapheresis are treatments for MS. While there is no cure, there are treatments that slow the progression of ALS. Riluzole is a Glutamate antagonist, which reduces excitotoxicity and is modestly effective in slowing the progress of ALS. Edaravone is an antioxidant, and also modestly effective in slowing ALS progression.
3. C - The weakness associated with ALS is localized to the Anterior Horn Cells.
4. A - hypotonia without associated weakness in an infant is a sign of upper motor neuron localization. All other findings presented are also consistent with this. By the time the child with an upper motor neuron issue becomes 2-3 years old, the hypotonia will transition to hypertonia.
5. C - Cerebral palsy is primarily diagnosed through a physical examination, during which a doctor will observe the individual's movements and muscle tone, and may also conduct tests of their reflexes and sensory abilities. CP can only be diagnosed after the patient is 2 years old. The physical examination may be supplemented with imaging tests, such as an MRI or CT scan, to assess the brain and spinal cord for any structural abnormalities. However, the physical examination is the primary method used to diagnose cerebral palsy.
6. A - Lack of oxygen to the brain, either *in utero*, or during or shortly after birth, is the most common cause of cerebral palsy. This can occur for various reasons, such as complications during labor and delivery, or a congenital heart defect that impairs blood flow to the brain. Possible risk factors for cerebral palsy include infections during pregnancy, traumatic brain injury, and genetic factors.
7. A - Spastic cerebral palsy is the most common type, accounting for about 70-80% of cases. This type of cerebral palsy is characterized by stiff and tight muscles, which can cause problems with movement and coordination. Other types of cerebral palsy include ataxic, athetoid, and mixed. Ataxic cerebral palsy is characterized by problems with balance and coordination, athetoid cerebral palsy is characterized by involuntary movements, and mixed cerebral palsy is a combination of two or more types.
8. D - All of the above. People with mitochondrial disorders may be at risk for a range of potential complications, including seizures, difficulty swallowing, and heart problems. These complications can vary depending on the specific type and severity of the disorder, and may require treatment to manage.
9. A, C, and D - B is false, as CSF lactate is likely to be high with mitochondrial disorders
10. C - The patient's symptoms, family history, and genetic test results are all consistent with a diagnosis of Huntington's disease. This is a neurodegenerative disorder caused by a mutation in

the HTT gene, which leads to the death of nerve cells in the brain and the symptoms of the condition. In contrast, Alzheimer's disease is a disorder that affects memory and cognitive function, Parkinson's disease affects movement, and ALS affects muscle function.

11. B - 40 is the cutoff for being diagnostic of Huntington Disease.
12. A, B, C, and E - Papilledema is a late response. Brain herniation is a terminal stage of increased ICP, resulting in shifting of the brain from one compartment to another, cranial nerve palsies, posterior cerebral artery compression or infarction, and possibly death.
13. D - Anaplastic glioma is grade III, and has three subtypes: Astrocytoma, Oligodendroglioma, and Ependymoma. Pilocytic astrocytoma is grade I, Low grade glioma is grade II, and Glioblastoma multiforme is grade IV.
14. B- Pramipexole is a dopamine agonist, a first-line treatment in early Parkinson's disease
15. C
16. D - H. influenzae type B, is a gram negative coccobacilli that can cause meningitis symptoms in infants, especially those too young for the vaccination.
17. D- for an immunocompetent adult with meningitis the appropriate medication regime is IV Ceftriaxone, Vancomycin, and Dexamethasone
18. A - Synkinesis is when the axon regenerates and attaches to a new target for example, the obicularis oris from a nerve that originally supplied the obicularis oculi.
19. D- Decreased vibration sensation is a key symptom in vitamin B12 deficiency, given the Chron's history and the clinical picture it is the most likely diagnosis and the expected exam finding.
20. D - Previous strokes can lead to dysphagia and increase the risk for aspiration pneumonia. Given the clinical picture of aspiration pneumonia her most relevant risk factor is the previous stroke.